

Fertility, Baby Booms, and the Political Economy of Housing Supply

Draft for project 03
(extension of the NIMBY housing-supply model)

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Abstract

This paper develops a fertility extension of the NIMBY housing-supply framework from project 02. The baseline project-02 mechanism links local political preferences to endogenous housing supply tightness. The extension adds endogenous fertility and a lagged cohort-political channel: a temporary baby-boom shock can increase later political resistance to housing supply, raise future housing prices, and reduce later fertility. We present side-by-side simulations against an old-model proxy that excludes the fertility channel, and we clarify exactly what the current prototype can and cannot establish. The model is a mechanism exercise and does not yet identify causal effects in US data. We close with an empirical implementation plan for a US fertility panel with explicit immigration/cohort accounting.

1 Introduction

The main question is whether demographic shocks can feed back into housing politics and then affect fertility in the next generation. In the original NIMBY model, housing supply outcomes are determined by political equilibrium and demand pressures. In this extension, fertility and demographics are no longer exogenous background objects: they are part of the state transition.

The conceptual sequence is:

1. A baby-boom shock temporarily raises births.
2. As these cohorts age into politically relevant groups, local supply preferences tighten.
3. Tighter supply raises housing costs and can lower subsequent fertility.

This paper is intentionally separate from project 02. It reuses the political housing-supply core, but asks a distinct question and introduces new dynamic channels.

2 Relation to the NIMBY paper

Table 1 summarizes what is inherited from project 02 and what is new here.

3 Model

3.1 Environment

Time is discrete. The state includes age-distribution mass $M_t(j)$, housing price index p_t , political tightness θ_t , and lagged births that proxy children-at-home demand pressure.

Households in fertile-age bins choose fertility subject to housing costs. Housing supply depends on political tightness and prices, as in the project-02 logic.

Table 1: Model crosswalk: project 02 versus project 03

Component	Project 02 (NIMBY paper)	Project 03 (this paper)
Political housing-supply block	Core mechanism	Inherited
Housing price dynamics	Endogenous	Inherited
Fertility choice	Exogenous / absent	Endogenous in new model
Demographic feedback to politics	Limited / exogenous demographics	Explicit lagged baby-boom cohort channel
Main experiment	Housing-policy and demand shocks	Temporary baby-boom shock plus fertility channel
Empirical target	Housing outcomes and political economy	Fertility dynamics plus housing politics

3.2 Fertility block (new channel)

Desired fertility rate:

$$f_t^* = \text{clip}(f_{level} \exp(-\eta_p p_t), 0.05, 0.85).$$

Observed births with a temporary baby-boom multiplier μ_t :

$$b_t = f_t^* \cdot \sum_{j \in \mathcal{F}} M_t(j) \cdot \mu_t, \quad \mu_t = \begin{cases} 1 + \text{boom_amp}, & t \in [t_{start}, t_{end}] \\ 1, & \text{otherwise.} \end{cases}$$

3.3 Political tightness

$$\theta_t = \text{clip}(\theta_0 + \theta_{old} \cdot old_t - \theta_{young} \cdot young_t + \theta_{boom} \cdot \bar{b}_{t-\ell:t-\ell-w+1}, 0, 0.95),$$

where the lagged births term captures entry of baby-boom cohorts into politically relevant homeowner/voter groups.

3.4 Housing market update

Demand:

$$D_t = d_0 + d_{young} \cdot young_t + d_{birth} \cdot b_t + 0.15 \cdot n_t^{home}.$$

Supply:

$$S_t = s_0 + \varepsilon_0(1 - \theta_t) \exp(-p_t).$$

Price law of motion:

$$p_{t+1} = (1 - \lambda)p_t + \lambda[p_t + g(D_t - S_t)].$$

4 Simulation design

Baseline calibration (current prototype):

- Horizon $T = 80$.
- Damping $\lambda = 0.25$.
- Baby-boom shock: +60% births from $t = 15$ to $t = 24$.

- Post-shock evaluation window starts at $t = 40$.

We compare:

1. **New model**: endogenous fertility and lagged boom-to-politics channel.
2. **Old proxy**: pre-extension logic with no endogenous fertility response.

5 Side-by-side simulation results

Figure 1 and Figure 2 provide dynamic paths and shock-minus-baseline effects.

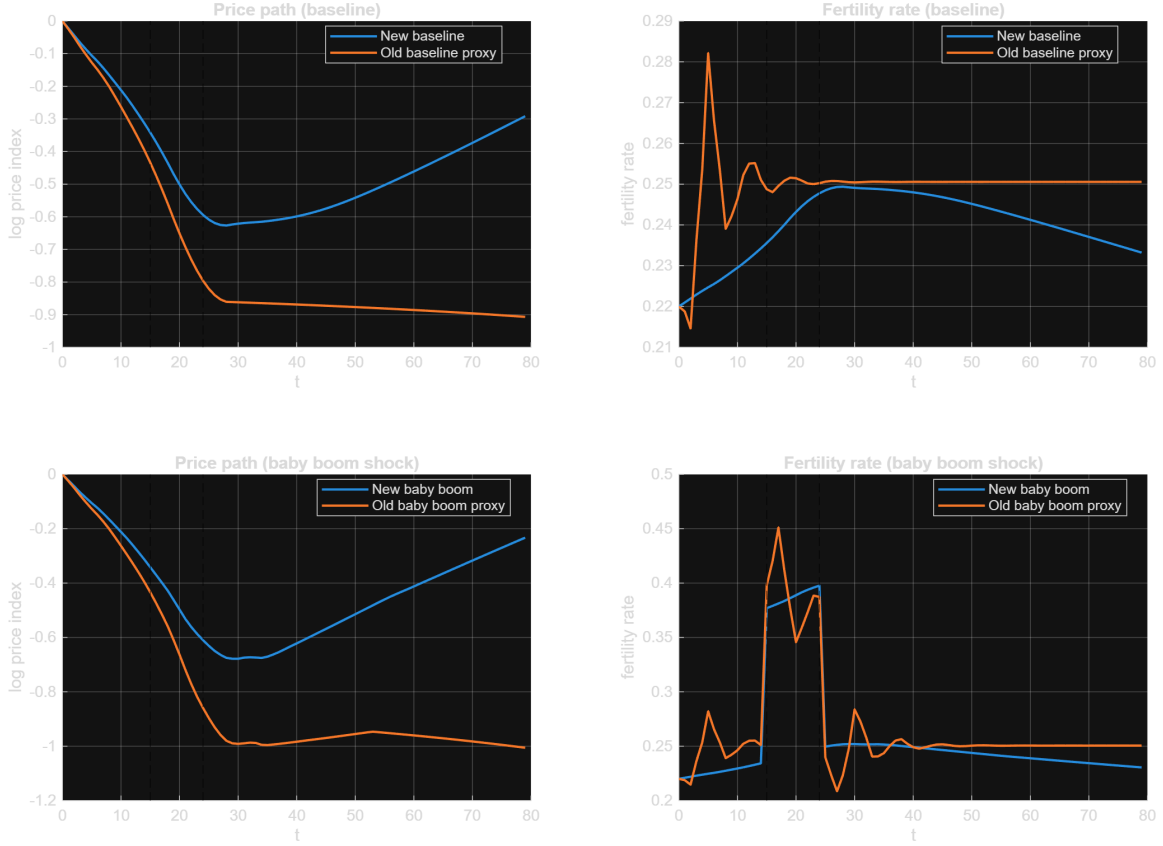


Figure 1: Baseline and shock paths, new model versus old proxy

Table 2 summarizes the key phase deltas from the latest run.

Table 2: Phase deltas (shock minus baseline) from latest MATLAB run						
Model	Δf_{pre}	Δf_{boom}	Δf_{post}	Δp_{pre}	Δp_{boom}	Δp_{post}
New	0.000000	0.145361	-0.001812	0.000000	-0.001259	0.038142
Old proxy	0.000000	0.141054	-0.000107	0.000000	-0.019750	-0.085626

Interpretation:

1. During the boom window, fertility rises mechanically in both models.

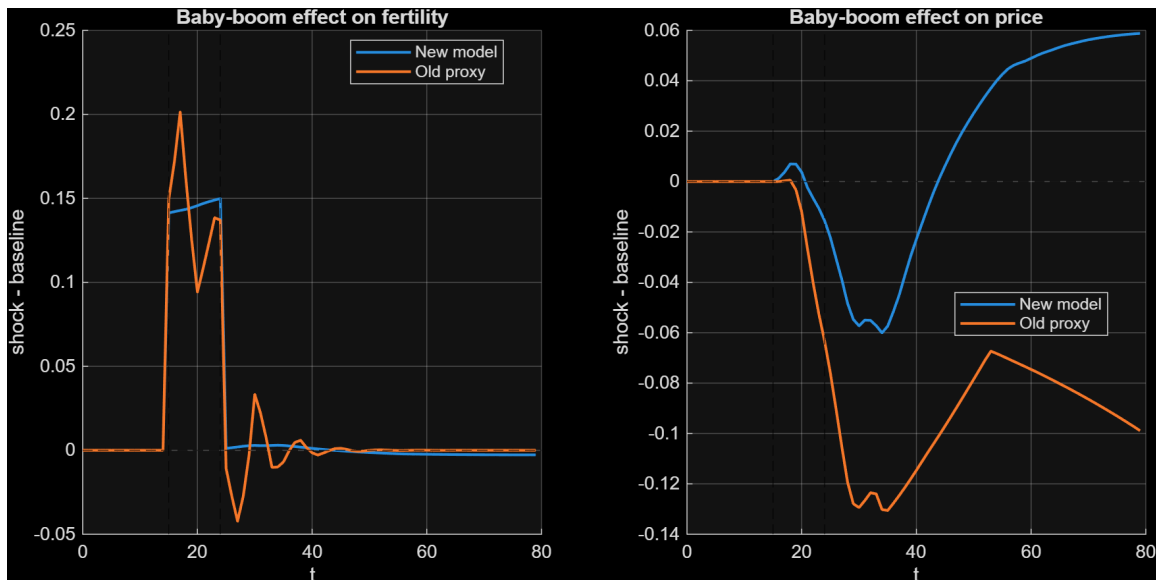


Figure 2: Shock-minus-baseline effects on fertility and prices

2. In the new model, post-shock prices are higher and post-shock fertility is lower than baseline.
3. The old proxy does not deliver the same post-shock direction, highlighting the added channel.

6 What this does and does not show

6.1 What it shows

The extension is internally consistent with the mechanism: baby-boom cohorts can tighten later housing politics and modestly reduce later fertility, conditional on calibration.

6.2 What it does not show

1. It does not yet establish a causal empirical effect in US data.
2. It is not yet the full project-02 steady-state stack; required upstream `.mat` inputs are not present in this repository copy.
3. Immigration and migration composition effects are not yet estimated empirically.

7 Empirical implementation: US fertility panel

The immediate empirical objective is to test whether the model-implied dynamic pattern appears in US data once migration/immigration composition is handled explicitly.

Panel target:

- Geography-time: county-year or MSA-year panel (state-year fallback).
- Outcomes: age-specific fertility rates, first-birth timing proxies, cohort-completed fertility proxies.

- Housing variables: permits, completions/unit mix, price/rent indices, reform timing.
- Demographic controls: migration flows and foreign-born share changes by age cohort.

Identification:

1. Staggered event study around housing-supply reforms.
2. Political close-election design where feasible.
3. Geographic-supply IV as complementary margin.

All baseline specifications will include immigration/cohort decomposition terms so that observed fertility changes are not conflated with compositional shifts.

8 Conclusion

This paper is a separate project built on the NIMBY framework with one core extension: endogenous fertility and lagged demographic-political feedback. The current model delivers a plausible mechanism in which temporary baby booms can worsen later housing-supply politics and reduce subsequent fertility. The next stage is empirical discipline using a US panel with explicit immigration-aware cohort accounting.